

$$\begin{aligned}
 \langle \int g^A \int g^B \rangle &= \langle \int g^A \int \bigcirc \int g^B \rangle = \int^A \int^B \langle \int g^A \int \bigcirc \int g^B \rangle = \int^A \int^B \langle \int g^A \int \bigcirc \int g^B \rangle \\
 &= \int^A \int^B \langle \int g^A \int \bigcirc \int g^B \rangle
 \end{aligned}$$